

VANGUARD COLDCHAIN LOGISTICS - RFP MASTER PROCUREMENT SPECIFICATIONS

Contract Document Reference: RFP-2025-FLT-90 | **Planning Horizon:** 5 Years (60 Months) | **Fleet Size:** 90 Class 8 Tractors (30 units per hub)

1. Executive Procurement Brief & Operating Scope

This master agreement details commercial pricing, government incentive schedules, preventive maintenance rates, component overhaul liabilities, and guaranteed residual buyback terms for the four candidate Class 8 powertrain configurations evaluated for Vanguard ColdChain Logistics (VCCL). All proposals are binding for 90 tractor replacements and 120 matching multi-temperature refrigerated trailers across Hub-North (Allentown), Hub-Central (Greensboro), and Hub-South (Jacksonville). Annual discount rate for capital cash-flow discounting is established at **6.0%**.

2. Option 1: Battery Electric Powertrain Package (BEV_DEPOT)

OEM Supplier: ElectroHaul Dynamics (Model: E-Force 800)

Vehicle Capex: Tractor MSRP \$395,000 per unit. Fully integrated All-Electric Transport Refrigeration Unit (e-TRU) Trailer MSRP \$45,000 per unit.

Clean-Truck State Vouchers: Qualified under State Commercial Zero-Emission Truck Incentive Programs. Tiered allocation: \$75,000 per tractor for the first 15 tractors deployed per hub/state, and \$45,000 per tractor for units 16 through 30 in each hub.

Guaranteed 5-Year Residual Value: OEM buyback guarantee at Month 60 equals **22.0%** of initial gross tractor MSRP.

Preventive Maintenance (PM) Rate: \$0.085 per fleet mile (covers routine chassis inspection, brakes, electric drive fluid, cabin filtration).

Battery Pack & Operational Constraints: 550 kWh nominal battery pack with an active operating window of 85.0% usable depth-of-discharge (467.5 kWh usable capacity). En-route opportunity fast-charging is rated at 150 kW DC. If battery depletion requires en-route charging, transit delay averages 45 minutes per charge event. Delivery window SLA breach penalty equals \$1,200 per delayed load. Unmitigated battery exhaustion resulting in pharma cargo temperature excursions incurs a strict GDP cargo deductible penalty of \$15,000 per incident.

3. Option 2: Renewable Natural Gas / Bio-CNG Package (BIO_CNG)

OEM Supplier: BioPower Methane Motors (Model: TerraGas 12N)

Vehicle Capex: Tractor MSRP \$210,000 per unit. Mechanical-Drive TRU Trailer MSRP \$38,000 per unit.

Clean-Fuel Alternative Vouchers: Flat alternative-fuel incentive of \$20,000 per tractor across all 90 units.

Guaranteed 5-Year Residual Value: Buyback guarantee at Month 60 equals **20.0%** of initial gross tractor MSRP.

Preventive Maintenance (PM) Rate: \$0.178 per fleet mile (reflects spark-ignited gas engine oil change intervals, spark plug replacements every 45,000 miles, ignition coil packs, and exhaust catalyst maintenance).

DOT Cylinder Recertification: In accordance with Federal Motor Carrier Safety Regulations, Type-4 composite CNG fuel storage cylinders must undergo complete ultrasonic non-destructive testing (NDT) and hydro-structural recertification at Month 36 (Year 3) at a cost of **\$2,800 per tractor**.

4. Option 3: Clean Diesel & Solar-Hybrid TRU Package (DIESEL_HYB_TRU)

OEM Supplier: EcoPrime Technologies (Model: CleanHaul 2027 Hybrid)

Vehicle Capex: Ultra-Low Emission Clean Diesel Tractor (EPA 2027 / Euro VI-E compliant) MSRP \$178,000 per unit. Solar-Assisted Electric Hybrid TRU Trailer MSRP \$52,000 per unit.

Government Subsidies: \$0.00 (Ineligible for zero-emission clean truck vouchers).

Guaranteed 5-Year Residual Value: Buyback guarantee at Month 60 equals **26.0%** of initial gross tractor MSRP.

Preventive Maintenance (PM) Rate: \$0.112 per fleet mile (scheduled engine oil, DPF cleaning, DEF filter service, mechanical chassis).

Hybrid Refrigeration Architecture: The trailer features an 800W monocrystalline rooftop solar-PV array paired with a 480V 3-phase electric standby compressor. At the depot, trailers connect to shore power, eliminating 100% of yard idling fuel for pre-cooling. On the road during daylight hours, the rooftop solar array generates electric power offsetting **36.0%** of transit TRU electrical demand, with the remainder powered via an ultra-high-efficiency engine generator.

5. Option 4: Hydrogen Fuel Cell Powertrain Package (H2_FCEV)

OEM Supplier: AeroHydro Mobility (Model: HydroMax 300F)

Vehicle Capex: Tractor MSRP \$450,000 per unit. Cryo-Electric TRU Trailer MSRP \$50,000 per unit.

Clean-Truck State Vouchers: High-capacity ZEV incentive of \$80,000 flat per tractor across all 90 units.

Guaranteed 5-Year Residual Value: Buyback guarantee at Month 60 equals **18.0%** of initial gross tractor MSRP.

Preventive Maintenance (PM) Rate: \$0.135 per fleet mile (covers deionized coolant flush, air compressor filters, high-pressure hydrogen plumbing inspections).

Fuel Cell Stack Lifecycle Overhaul: The 300 kW PEM fuel cell stack requires a mandatory mid-life membrane electrode assembly (MEA) refurbishment and stack overhaul at 18,000 cumulative operating hours (scheduled at Month 48 / Year 4) at a fixed contract cost of **\$42,000 per tractor**.

6. Summary Comparison Matrix of Contract Terms

Strategy Token	Tractor MSRP	Trailer MSRP	Subsidy / Unit	PM \$/Mile	Year-5 Residual	Major Mid-Life Overhaul
BEV_DEPOT	\$395,000	\$45,000	Tiered (\$75k / \$45k)	\$0.085	22.0%	None (Covered under warranty)
BIO_CNG	\$210,000	\$38,000	\$20,000 flat	\$0.178	20.0%	Month 36 Tank NDT (\$2,800/unit)
DIESEL_HYB_TRU	\$178,000	\$52,000	\$0	\$0.112	26.0%	None (Standard PM)
H2_FCEV	\$450,000	\$50,000	\$80,000 flat	\$0.135	18.0%	Month 48 Stack Overhaul (\$42,000/unit)